ESP32

INTRODUCTION

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TOPIC OUTLINE

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ESP32 Module

ESP32 Dev Kit

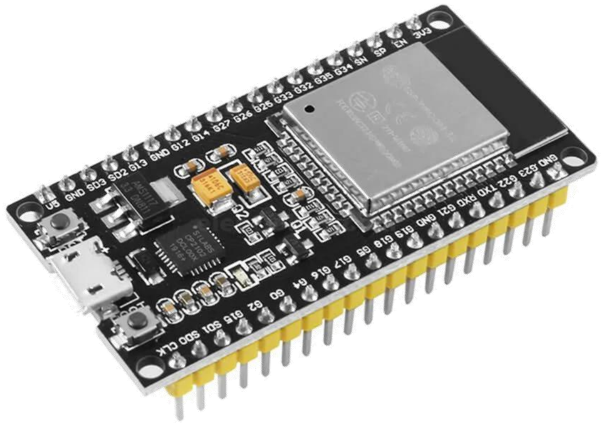
GPIOs

ESP32 VS ARDUINO

ARDUINO VS ESP32

The Arduino UNO is an excellent **beginner-friendly** microcontroller board designed for simple electronics and learning embedded systems.

The ESP32 is a much **more powerful microcontroller** with significantly higher processing speed, larger memory, built-in Wi-Fi and Bluetooth, and more input/output options.



MAIN DIFFERENCES

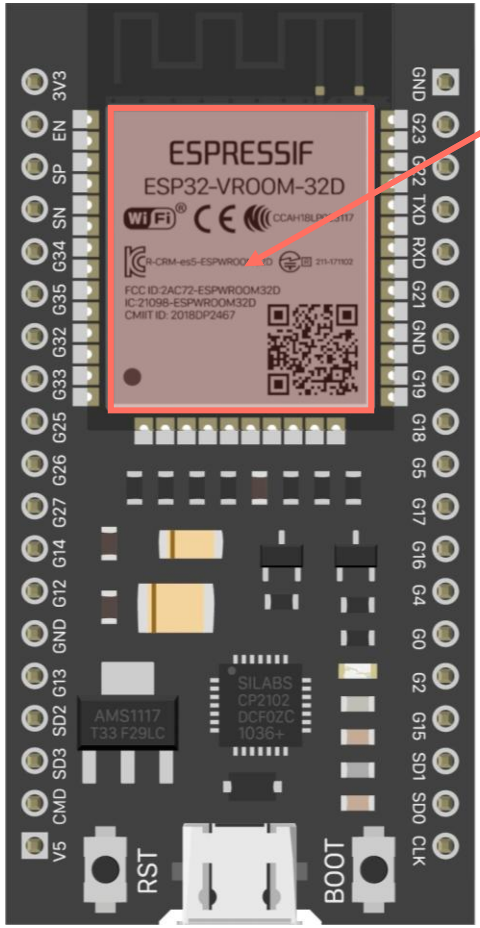
Feature	Arduino UNO	ESP32
Microcontroller	ATmega328P	Dual-core Xtensa LX6 (or similar ESP32 variants)
Clock Speed	16 MHz	Up to 240 MHz
Operating Voltage	5V	3.3V
Flash Memory	32 KB	Typically 4MB or more
RAM	2 KB SRAM	About 520 KB SRAM
Wi-Fi	Not built-in	Built-in
GPIO Pins	14 digital, 6 analog inputs	Up to 34 GPIO pins (varies by board)
Processing Power	Suitable for basic projects	Much faster for advanced applications
Power Consumption	Higher in active mode, simple sleep options	Advanced low-power modes for IoT devices
Programming	Arduino IDE	Arduino IDE, ESP-IDF, PlatformIO, etc.

ESP32 MODULE

ESP32 MODULE

The **ESP32** is a reference name to a variety of boards and modules based on the core ESP32 chip.

- ESP32-WROOM-32 module contains the ESP32-D0WDQ6 chip.
- ESP32-WROOM-32D module contains the ESP32-D0WD chip.
- ESP32-WROVER-IB module contains the ESP32-D0WD but with added PSRAM.

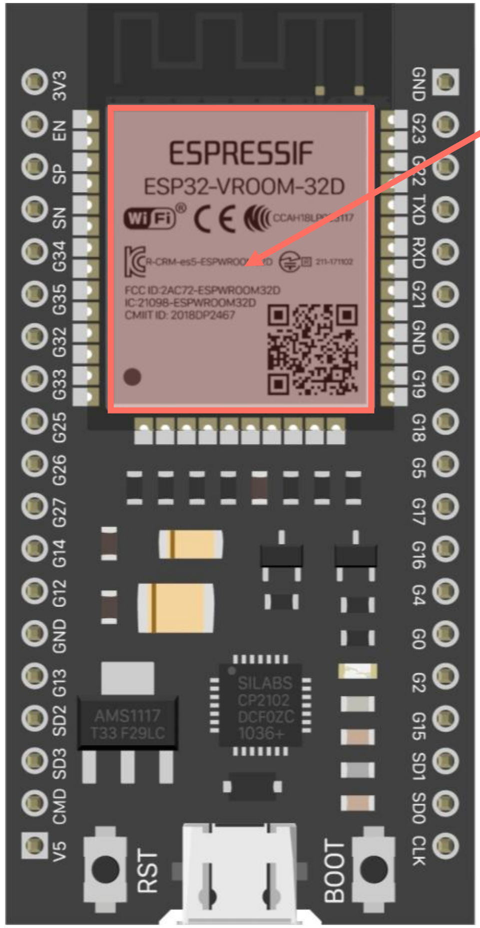


chip

ESP32 MODULE

Each module and chip combination has a unique set of characteristics.

- Amount of flash memory (typically **4 MB**)
- Presence and amount of **PSRAM** (pseudo-static RAM)
    - **8 MB** available in WROVER modules
- Type of antenna
    - **MIFA** (Meandered Inverted-F Antenna)
    - **U.FL** (connector for an external antenna)
- Number of processing cores
    - Chips with **D** after ESP32 denote **dual core**
    - Chips with **S** after ESP32 denote **single core**

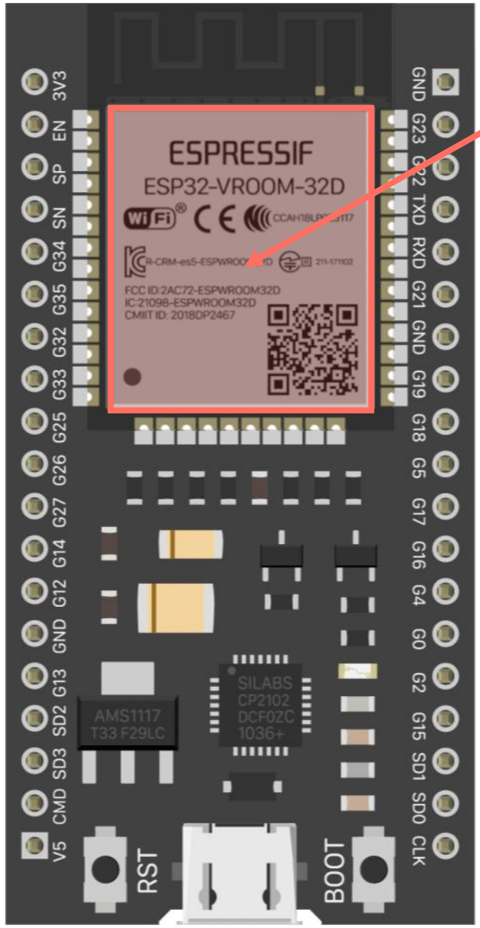


chip

ESP32-WROOM-32

The most commonly used module.

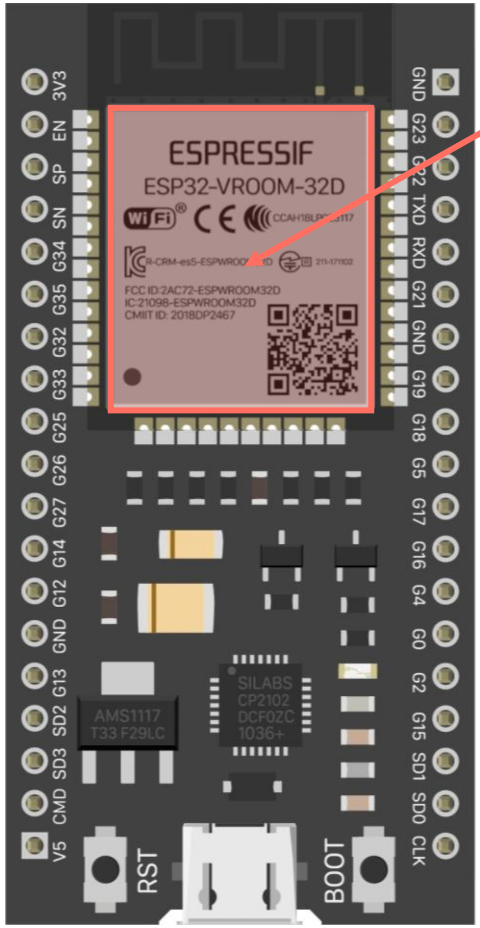
- Contains the **ESP32-D0WDQ6** chip
- **4 MB** Flash (some variants go up to 16MB)
- No PSRAM
- **MIFA** antenna



chip

ESP32-WROOM-32D/32U

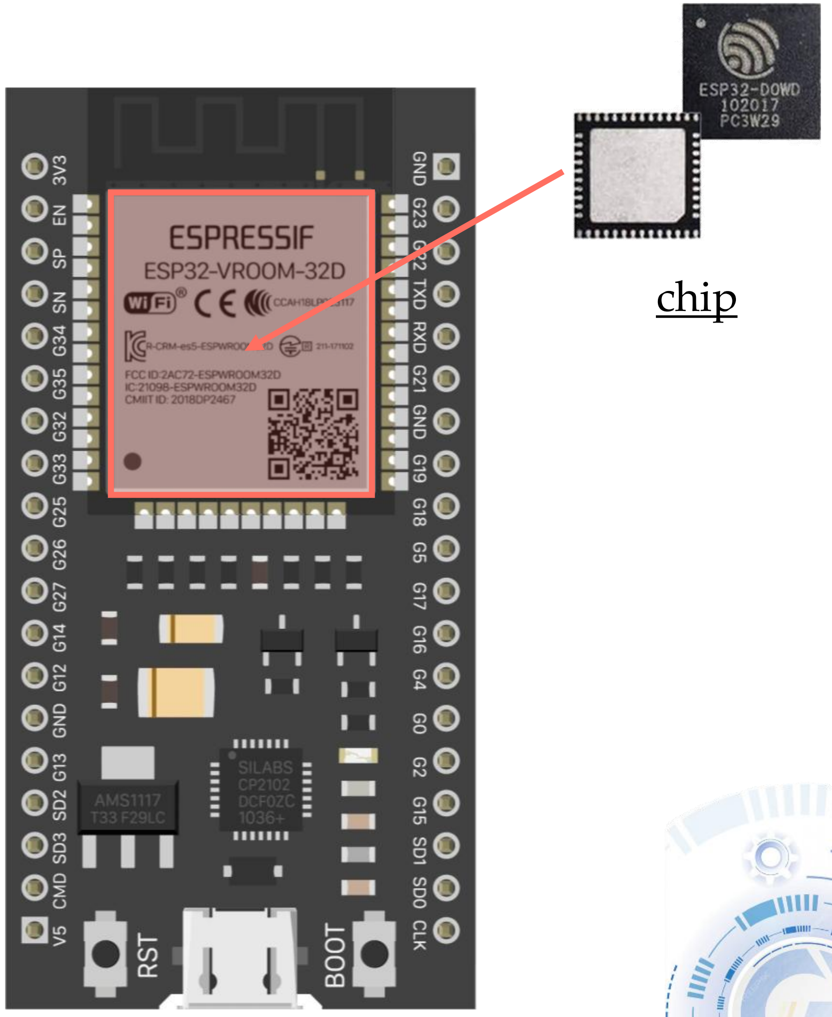
- Contains the **ESP32-D0WD** chip
- **4 MB Flash** (some variants go up to 16MB)
- No PSRAM
- **MIFA** antenna for the **D** model
- **U.FL** antenna connector for the **U** model
- Smaller footprint than the ESP32-WROOM-32



chip

ESP32-WROOM-32E/32UE

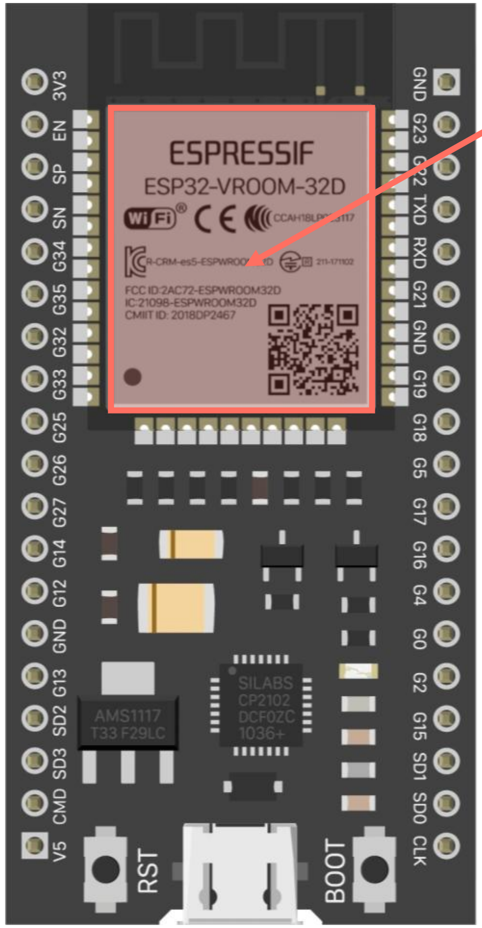
- Contains the **ESP32-D0WD-V3** chip, offering higher stability and safety performance compared to the ESP32-WROOM-32
- **4 MB** Flash (some variants go up to 16 MB)
- No PSRAM
- MIFA antenna connector for the UE model
- This is the module used in the official dev kit version 4.
- See https://espressif.com/en/products/modules for a listing of all current modules



ESP32-WROVER

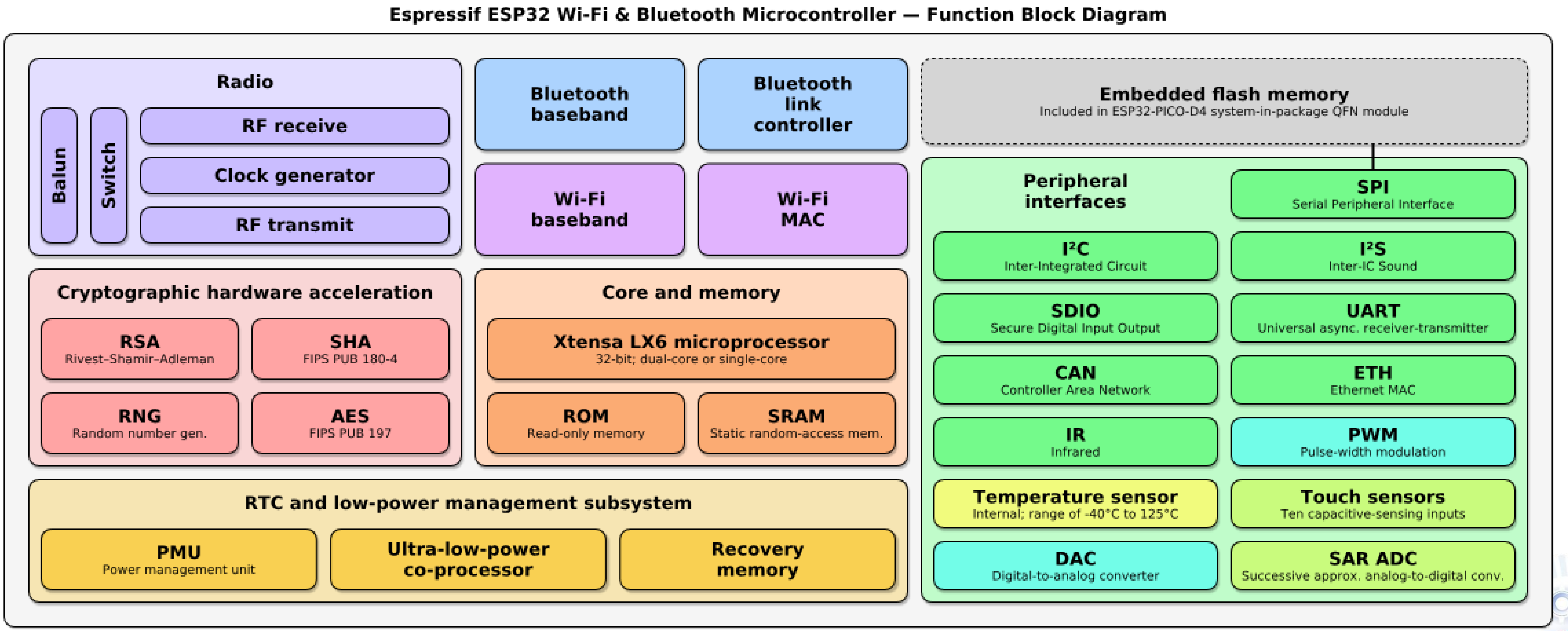
More powerful compared to the WROOM models

- **ESP32-WROVER** and **ESP32-WROVER-I** use the ESP32-D0WDQ6 chip (same as ESP32-WROOM-32)
- **ESP32-WROVER-B** and **ESP32-WROVER-IB** use the ESP32-D0WD chip (same as ESP32-WROOM-32D and U)
- **4 MB** Flash (similar to WROOM modules)
- **8 MB SPI PSRAM** (WROOM have none)
- **MIFA** or **U.FL** antenna
- Depending on the model, can operate at **1.8V**, and up to **144MHz** clock speed



chip

ESP32-D0WDQ6



ESP32 MODULE COMMON FEATURES

All ESP32 modules share these features:

- CPU cores (one or two)
- Internal memory (ROM, SRAM)
- External SRAM
- Timers and watchdogs
  - Four general-purpose 64-bit timers
  - Three watchdog timers (used to recover from faults)
- RTC clock
- 2.4 GHz receiver and transmitter radio
- Wifi, 802.11 b/g/n
- Bluetooth, classic and BLE
- RTC (co-processor) and Low-Power management with multiple power modes
- 34 GPIO pins
- Analog to Digital Converter (ADC)
- Hall sensor, capable to detect a magnetic field without additional hardware
- Digital to Analog Converter (DAC)
- Touch sensor via 10 capacitive sensing pins
- Ethernet MAC interface
- SD/SDIO/MMC host controller
- SDIO/SPI slave controller

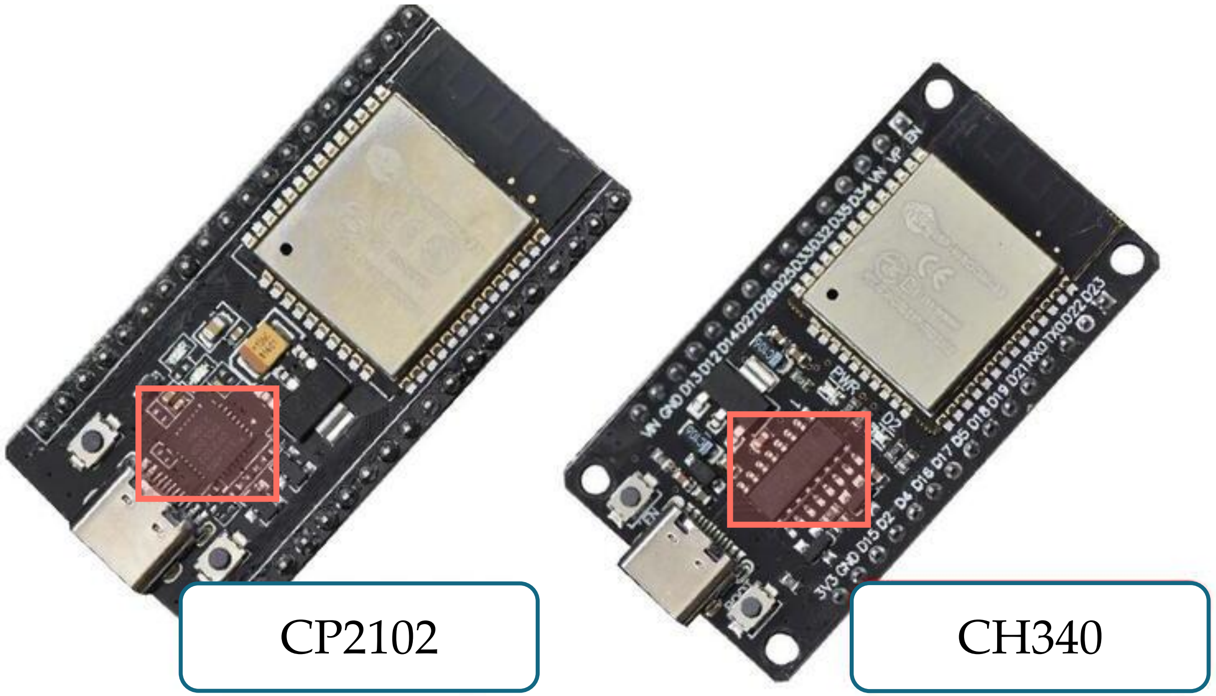
ESP32 MODULE COMMON FEATURES

All ESP32 modules share these features:

- UART
- I^{2}C
- Infrared Remote Controller
- Pulse Counter
- Pulse Width Modulation (PWM)
- LED PWM
- SPI
- Hardware acceleration of algorithms such as AES, RSA and ECC

ESP32 DEV KIT

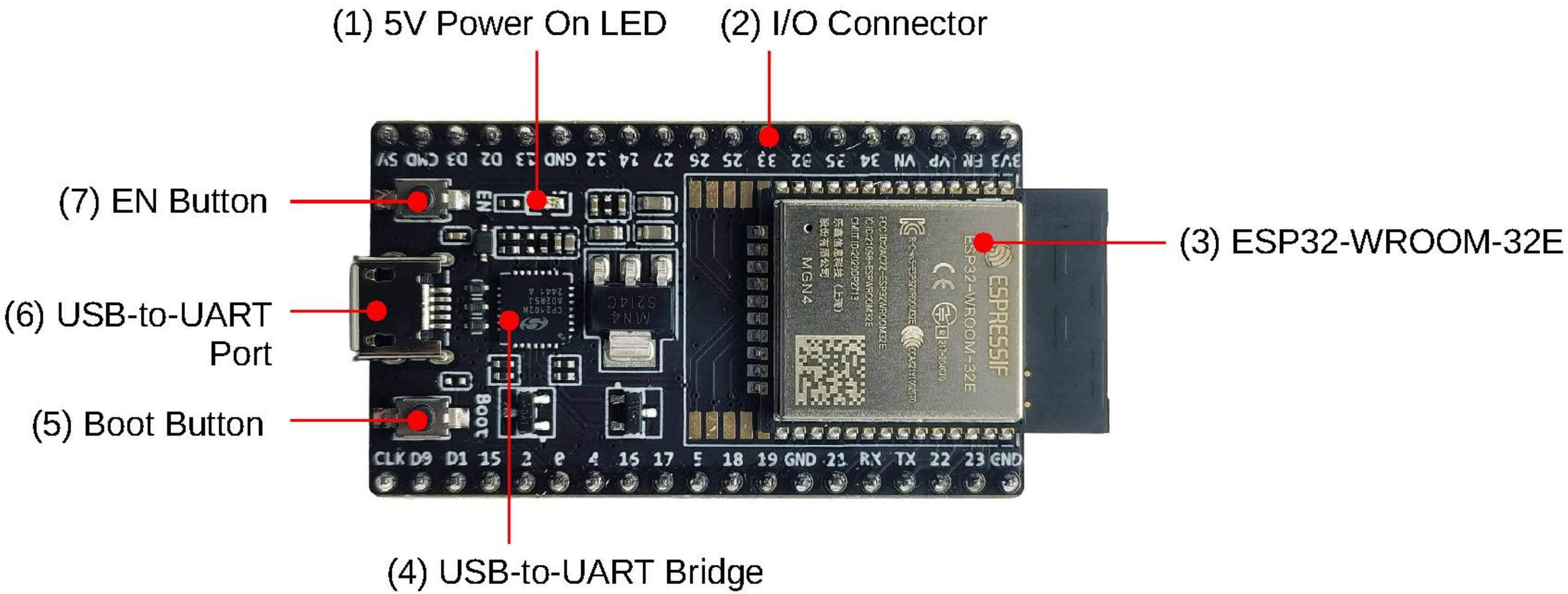
ESP32-DEVKITC V1



CP2102 and CH340 are USB-to-UART converter chips.

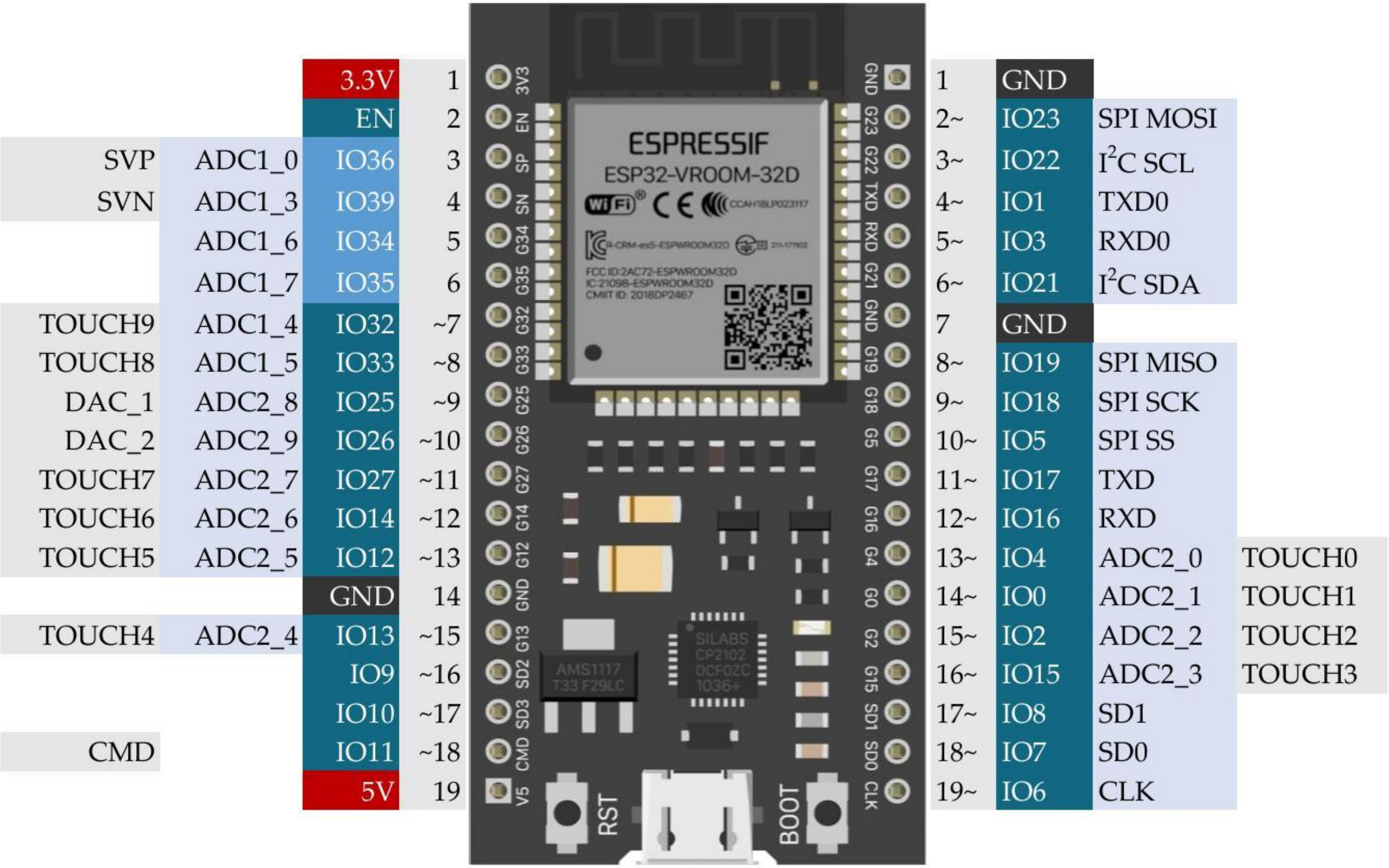
[CP210x USB to UART Bridge VCP Drivers - Silicon Labs]

ESP32-DEVKITC V4



GPIOs

DEVKITC GPIOs MAP



GPIO voltage: 3.3V (NOT 5V tolerant)

GPIO current: 12-20 mA (40 mA max)

ESP32-DevKitC-1 - - — Arduino ESP32 latest documentation

MATERIALS

Item	Price
ESP32 DevKitC WROOM-32D (38 pin)	385
ESP32 Expansion Board (38 pin)	130
Potentiometer (10K)	75
Photoresistor	25
IR Sensor Module	30
16x2 LCD Module with I2C	165
Logic Level Shifter	85
BME280 3.3V Environment Sensor	210
	1105

LABORATORY